

COMP526: Computational Methods for Scientists, Fall 2022

Assignment 8 - Due Wednesday 12/12/22

Under your home directory create directory assign8. In it create your python a source file assign8.py.

Use `scipy.integrate.odeint` to integrate numerically the differential equation

$$\frac{dy}{dt} = \cos t, \quad y(0) = 0,$$

for $t \in [t_0, t_f]$, with $t_0 = 0$ and $t_f = \frac{3\pi}{2}$.

- Assume initial step-size $h_0 = 10^{-7}$, and relative and absolute tolerances $rtol = 10^{-4}$, $atol = 10^{-7}$.
- Plot the numerical solution $\tilde{y}(t)$ and compare it with the exact solution $y(t) = \sin t$.
- Print the string "Solver rtol atol h0 mescd scd steps feval nJac CPU" and below it the following statistics:

1. Solver name
2. rtol
3. atol
4. h0
5. mescd

$$\text{mescd} = -\log_{10} \left(\frac{\|\tilde{y}(t_f) - y(t_f)\|_{\infty}}{\text{atol} + \text{rtol} \|y(t_f)\|_{\infty}} \right),$$

6. scd

$$\text{scd} = \begin{cases} -\log_{10} \left(\frac{\|\tilde{y}(t_f) - y(t_f)\|_{\infty}}{\|y(t_f)\|_{\infty}} \right), & \|y(t_f)\|_{\infty} > 10^{-12}, \\ -\log_{10} (\|\tilde{y}(t_f) - y(t_f)\|_{\infty}), & \|y(t_f)\|_{\infty} \leq 10^{-12}, \end{cases}$$

7. Cumulative number of time steps
8. Cumulative number of function evaluations per each time step
9. Cumulative number of jacobian evaluations per each time step
10. CPU time (compute this as the median of the CPU times of five runs of the integrator).

BONUS:

Produce the same table for one of the following University of Bari benchmark ODE systems (see the ODEBenchmarkTestSet.pdf file posted on Canvas). Use the same $rtol$, $atol$ and h_0 . I can provide the Jacobian as a MATLAB file, if needed. Please, select only one; once selected cannot change your choice.

1. HIRES (page II-1-1)
2. Van der Pol (page II-8-1)
3. Oregonator (page II-9-1)
4. Robertson (page II-10-1)